Beyond the Session: Evaluating Process-Level Supervision of Long-Horizon Agents through Workspace Evolution

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Abstract

Long-horizon agents work across context resets while persistently changing code, papers, data, experiments, and documentation. Final outcomes and session-local traces hide how workspace state evolved and which earlier changes a later session revisited or validated. Prior work diagnoses trajectories and injects guidance, but does not test structured workspace evidence against complete same-source records exposed by Full Raw Retrieval (Full Raw). We hypothesize that workspace relations make relevant cross-session evidence accessible within a bounded budget and improve an automatic supervisor's executed continuation. We present AGENT NEBULA, a deterministic, source-linked workspace action trajectory and bounded retrieval interface. At a frozen checkpoint, we define a comparison among No-op, Generic, Full Raw, and Workspace Trajectory. A supervisor intervenes or abstains, the same worker continues from byte-identical forks, and the unmodified official executable evaluator (oracle) scores every result without human annotations, agent-generated labels, or LLM-judge scores. A mechanics pilot had zero historical-evidence calls, and a preregistered six-task No-op headroom screen blocked the planned four-condition objective-utility comparison. We report no treatment-effect estimate and specify the required prospective cross-domain qualification.

Introduction

Long-horizon agents accept broad goals, work for hours or days, and return a repository or research workspace after many model contexts. Software development and autonomous research share this structure because agents read, create, modify, rename, validate, and delete persistent artifacts while their internal context is repeatedly replaced. Persistent multi-artifact workspaces already appear in realistic benchmarks (Zhou et al. 2026).

A session is therefore an incomplete unit of oversight. It records where an action executed, but the workspace preserves what one session leaves for later sessions to revisit or replace. Final task scores are necessary but also collapse the process. Two workers can reach the same score after different degrees of rediscovery, unvalidated change, or discarded work. Context management can drop task-critical plans (Mehta and Datta 2026), and failures can emerge only over many sessions (Zhu et al. 2026a). Linear logs preserve more evidence, yet force a bounded supervisor to reconstruct artifact lifecycles and cross-session continuity inside its context window.

Existing diagnosis and intervention methods do not establish objective intervention utility under a matched Full Raw–Workspace Trajectory comparison. We define this utility as improvement in the official outcome of the subsequently executed continuation under matched resources. AgentDiagnose measures process competencies (Ou et al. 2025), while AgentRx localizes critical steps using synthesized constraints and stepwise validation logs (Barke et al. 2026). TrajAudit retrieves evidence from noisy repository traces (Wang, Xie, and Huo 2026), and AgentForesight audits prefixes for early failure (Zhang et al. 2026). Similarly, intervention alone is not the contribution: REFLECT validates attribution through controlled replay (Lin et al. 2026), while harness-oriented work links trace mechanisms to repairs (Chen et al. 2026).

The narrower open question is whether source-linked cross-session workspace relations improve objective intervention utility beyond the same bounded supervisor with complete same-source Full Raw access. In this comparison, Full Raw is complete with respect to every frozen item in the registered Raw universe, including sanitized native sessions, prior official prompts and worker-visible logs, and immutable checkpoint or snapshot bytes and manifests. Completeness is limited to those records and does not imply observation of effects absent from them.

Our hypothesis is that persistent-workspace relations can make relevant cross-session evidence accessible within a bounded retrieval budget and thereby improve the realized outcome of a later worker. The hypothesis does not claim that the relations add source information, and trace readability alone does not establish it. Support requires executed continuations to outperform both complete Full Raw access and matched extra inference.

We present AGENT NEBULA to test that hypothesis without semantic diagnosis labels. It orders native actions by normalized reported execution time, projects only source-supported artifact effects, and exposes artifact history, differences between immutable post-session snapshots, and whether each action's effects are observed, absent under qualified coverage, or unresolved. Both Full Raw and Workspace Trajectory draw on the same registered Raw universe. At a frozen checkpoint, each bounded supervisor emits one guidance message or ABSTAIN. Every fork then executes the same future task with the same worker. The oracle measures the realized continuation outcome. No-op and Generic control for unassisted continuation and extra inference. This closed loop avoids asking either a person or another model to decide whether the process "looked wrong."

The current implementation establishes mechanics but not objective intervention utility. Its pilot and headroom results remain dependency-only findings.

This paper makes three contributions:

1. A workspace-centered formulation of process-level supervision across sessions and artifact types, evaluated by objective continuation outcome (Section).
2. A deterministic, source-linked action trajectory and bounded query interface exposing artifact history, session differences, and action effects while preserving exact-record provenance and source-and-budget parity with Full Raw (Section).
3. A checkpoint-matched intervention protocol designed to distinguish objective intervention utility from extra inference, worker variation, task headroom, and evaluator leakage using No-op, Generic, and Full Raw controls (Section).

Problem Setting and Motivation

Persistent Workspaces and Checkpoints

Let a long-horizon task operate on a persistent workspace W through model sessions $s_1, \ldots, s_m$. A session may end because of context limits, interruption, delegation, or orchestration policy while both task and workspace continue. Let C be a structural checkpoint immediately before a fresh top-level worker session. It freezes the complete workspace bytes, admitted source records, task state, worker configuration, and remaining official continuation. The checkpoint, rather than a completed conversation, anchors the comparison in Section .

The Workspace-Centered Evidence Gap

The final workspace W_T cannot distinguish processes that leave similar artifacts. A native transcript retains detail but partitions continuing work by transient contexts. A workspace-centered view connects actions through durable artifacts across sessions. This representation does not prove intent or causality. It makes the observed cross-session action sequence retrievable under a bounded supervisor budget.

The claim is deliberately comparative. Workspace Trajectory contains no native fact unavailable through Full Raw Retrieval. Its hypothesized value is an inductive bias: artifact history, session differences, and action-effect status may make useful evidence accessible before the retrieval budget is exhausted. For example, a configuration may be renamed in one session, followed by a test edit and no observed validation in later sessions. Full Raw access requires the supervisor to discover both names and open several ranges. Artifact history, session-difference and effect queries can return supporting source IDs in bounded requests. This is a motivating access pattern, not evidence of utility. If Full Raw matches Workspace Trajectory's continuation outcomes, the objective intervention utility claim fails. A tie that meets a preregistered lower-retrieval-cost criterion supports only an efficiency claim.

Supervision Target

The evaluated consumer is an automatic supervisor, not a human visualization user. It may inspect checkpoint evidence, emit one bounded intervention, or abstain. The intervention is useful only if the subsequently executed worker receives a better official outcome without unacceptable harm or cost.

Workspace Trajectory Design

Overview and Requirements

Four requirements constrain the design. *Continuity* requires a common ordering across sessions and source adapters that have passed source/effect qualification. *Evidentiary fidelity* requires the projection to separate observed effects from inferred semantics and link every relation to exact source records. *Budgeted access* requires retrieval from long histories under fixed byte, token, tool, and time budgets. *Fair comparison* requires Full Raw and Workspace Trajectory to have identical source membership and total supervision and continuation resources. Within qualified adapters, continuity and fidelity are checked through ordering, provenance, and parity. Budget and fairness are protocol controls. Only objective intervention utility is an empirical claim. AGENT NEBULA normalizes reported native timestamps to epoch time, and the source ID deterministically orders equal timestamps. Records without a usable timestamp remain available to Full Raw lookup but do not create ordered transition edges. Git commits do not define the action time axis or establish progress. They may only mark milestones in an auxiliary replay.

Figure 1 shows the matched comparison. Both conditions begin with the same frozen native records. Full Raw exposes bounded search, exact-record, and exact-range access. Workspace Trajectory retains those tools and adds only deterministic relations derived from the same records. A shared supervisor and worker produce an executed continuation, which the official oracle scores.

Source-Linked Workspace Trajectory

Projecting only files would erase searches, failed commands, and validation actions. We therefore retain every native tool action and attach zero or more source-supported artifact effects. An action is

$$a_i = (t_i, s_i, v_i, c_i, u_i, q_i), \tag{1}$$

where t_i is the native timestamp, s_i the session, v_i the vendor, c_i the native call identifier, u_i the native tool/category, and q_i its reported success, failure, or unknown call status. The source ID is $id_i = (s_i, c_i)$. An action-effect status $h_i \in \{\textsf{observed}, \textsf{no\_effect}, \textsf{unknown}\}$ records whether qualified source evidence resolves the action's workspace effects. Read-only lookup $R(id_i)$ returns the frozen native arguments and result. A deterministic projection associates a_i with

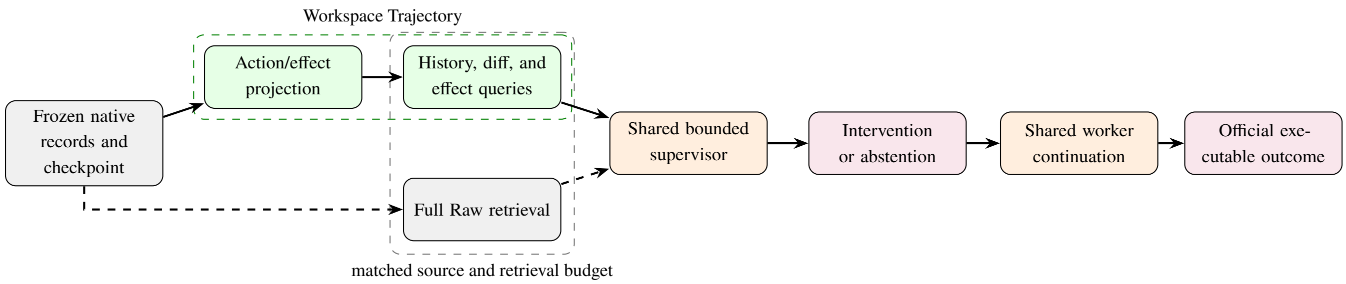

Figure 1: Matched comparison of Full Raw and Workspace Trajectory. Both use the same frozen sources and retrieval budgets; only Trajectory adds deterministic relations. The official oracle scores the same worker's continuation after a bounded intervention or abstention.

$$E_i = \{(p, o, p')\},\quad o \in \{\textsf{read},\textsf{write},\textsf{create},\textsf{delete},\textsf{rename}\}, \tag{2}$$

where p is workspace-relative and p' is the previous path only for an observed rename. A failed call contributes no successful effect. no_effect is assigned only when the qualified adapter has complete declared effect coverage for the action and observes no effect. Missing or ambiguous evidence remains unknown. Both no_effect and unknown may have $E_i = \emptyset$, but the index preserves their distinction and keeps unknown candidates retrievable. Any unresolved disagreement about the task or workspace boundary blocks supervisor execution. The trajectory is

$$\mathcal{T}(W, C) = \mathrm{sort}_{(t_i, id_i)}[(a_i, h_i, E_i)]. \tag{3}$$

The source ID serializes as `session-id#source-call-id`. Full Raw resolves it to the frozen native record, and Workspace Trajectory preserves it on every derived relation. A trajectory-only ordinal is not a source ID.

Lifecycle and snapshot transitions. For each artifact, the index maintains names, observation bounds, last access, read/write counts, and explicit lifecycle events. A frozen checkpoint manifest marks files present at C. Otherwise, existence before first observation is unknown. Native rename preserves identity, but delete followed by create does not unless the source explicitly connects them.

Let B_s be the immutable manifest of the post-session workspace snapshot. The session difference $D(B_s, B_{s'})$ contains exact added, removed, and content-changed artifacts between consecutive admitted snapshots. It is a state difference, not a causal dependency. Directory hierarchy remains presentation metadata rather than an additional file-like object.

Bounded Supervision Interface

All supervisor arms can list, search, and read the current workspace. Full Raw adds a four-operation read-only interface over the registered Raw universe. The `list_sources`, `search`, `read_record`, and `read_range` operations expose source membership, lexical matches, exact records or snapshot bytes, and contiguous source intervals.

Workspace Trajectory receives the same Raw tools plus exactly three deterministic relation queries. `artifact_history(path)` returns ordered source-backed reads, mutations, versions, renames, and deletion. `session_diff(from,to)` returns exact added, removed, or changed artifacts between immutable post-session snapshots. `effects(action)` returns the action-effect status, observed effects, unresolved candidates, and supporting source IDs. The registered interface exposes no hotspot, importance, recurrence, validation, anomaly, or semantic query.

Queries return observations and provenance, not intent, failure reasons, or diagnostic labels. The shared retrieval layer charges the budget for rendered requests, schemas, arguments, returned bytes and tokens, conversation history, and responses. The comparison is therefore about bounded accessibility and inductive bias, not new source information.

Intervention policy. The supervisor prompt fixes one output schema: a bounded guidance message or `ABSTAIN`, cited source IDs when evidence is used, and a short action rationale. It cannot write the workspace, invoke the worker's hidden tools, or inspect future prompts and evaluator fixtures. The message is appended to the otherwise unchanged next official task. Every call and cited source is logged for the historical-evidence engagement gate in the evaluation.

Implementation

Agent Nebula is implemented in Rust using the repository's existing parser for native agent sessions. It affiliates sessions with a workspace, orders tool calls by native time, and resolves workspace-relative paths. The research harness adds read-only retrieval, checkpoint cloning, supervisor/worker adapters, budget accounting, and oracle invocation without defining a second event schema. The implementation parses and visualizes Claude, Codex, and Gemini native records. The completed research qualification, however, covers exact source/effect binding only for Codex.

Source Ingestion and Provenance

Directly affiliated sessions are read completely. Paths are resolved against known worktree roots, and records outside the workspace remain source actions without invented repository effects. Dependency and build trees such as `.git`, `node_modules`, `target`, and `.cache` are excluded from the repository projection. Failed calls remain chronological actions. Missing or ambiguous paths close as unknown, not no_effect.

Every projected relation retains its source ID. Checkpoint creation freezes the workspace manifest, admitted session IDs and byte hashes, task boundary, worker argv/environment, future prompt identity, and remaining budget. Concurrent records use (t_i, id_i) for reproducible order without claiming causality.

Matched Retrieval and Continuation

Full Raw and Workspace Trajectory share one source store and retrieval layer. The retrieval layer enforces returned-byte, rendered-token, tool-call, response, time, and cost ceilings at the interface rather than through post-hoc line counts. It records all queries, responses, source IDs, tokens, latency, and supervisor output. The harness clones the frozen workspace once per condition, injects at most one bounded message, and launches the same worker command on the same official next task.

The completed run invoked the oracle twice, rejected evaluations that disagreed, and retained only the first payload and invocation count. A post-run code repair configures future runs to retain both payloads and their joint hash. The inspected tasks were not rerun after this repair. The completed run also mistakenly archived runtime credential state, which was removed without reading. A separate post-run repair configures future executions to purge credential and session runtime trees. Publication excludes all such paths.

The auxiliary AGENT NEBULA replay uses action time rather than commit time and exports the same projection for inspection. Neither visual aesthetics nor human reading speed provides evidence for the automatic-supervision RQs.

Evaluation

RQ1: Objective Intervention Utility. From the same frozen checkpoint, does Workspace Trajectory improve the official continuation score relative to Full Raw, and does its gain over No-op exceed Generic's gain, under fixed supervisor, worker, and total resource budgets while satisfying the harm criteria versus No-op?

RQ2: Query Contribution. Only if RQ1 is supported do we ask two questions: (1) does removing artifact history, session differences, or action effects reduce objective intervention utility, and (2) is any gain attributable only to earlier-session source access?

RQ3: Generalization and Safe Supervision. Conditional on an RQ1-supported Workspace Trajectory advantage, does that advantage generalize across held-out coding and scientific-work workspace/task families, workers, and models while satisfying the preregistered intervention-harm, abstention, and budget criteria?

Prospective Experimental Protocol and Analysis

The experimental unit is (C, F), where C denotes the frozen checkpoint and F denotes the unchanged future task and continuation budget. Supervisor policy π_k observes only condition-k evidence available at C and returns bounded guidance z_k or `ABSTAIN`. The fixed worker A executes

$$W'_k = A(C, F, z_k), \qquad y_k = O(W'_k), \tag{4}$$

where O is the unmodified executable oracle. The experiment executes all four continuations. No model estimates a counterfactual outcome.

We instantiate each unit at a structurally selected checkpoint before a fresh top-level worker session. Earlier official rounds create the prefix workspace and source history. Before condition execution, the harness freezes all inputs and clones four byte-identical forks:

1. **No Intervention (No-op)**: unchanged official continuation.
2. **Generic**: matched reflection/inspection guidance without historical evidence access.
3. **Full Raw**: bounded search/read access over the complete registered Raw universe.
4. **Workspace Trajectory**: the same Raw tools and budget plus the deterministic workspace relations in Section .

All four forks share the checkpoint, future task, worker configuration, worker continuation budget, and oracle. No-op invokes no supervisor. Generic, Full Raw, and Workspace Trajectory share the supervisor model/version, base prompt, output schema, decoding configuration, and budget ceilings, but receive condition-specific evidence interfaces. Generic has no historical-evidence tools. Full Raw has Raw search/read. Workspace Trajectory has the same Raw tools plus relation queries. Condition-specific schemas and returned evidence are charged to the same ceiling. Condition order is randomized within checkpoint. Related repositories, task variants, and workspace histories remain in one held-out cluster. Eligibility and checkpoint rules are frozen before observing scores.

The supervisor cannot see future prompts, hidden fixtures, evaluator code or results, repaired siblings, or outputs from another condition. Every relation visible to Workspace Trajectory cites a record retrievable by Full Raw. The matched engagement gate requires one successful condition-specific call: a current-workspace inspection for Generic, a Raw-history call for Full Raw, and a registered relation query for Workspace Trajectory. The returned payload must expose at least one registered source ID; an empty successful API response does not count. A condition that misses its family cannot satisfy the protocol.

Measures and interpretation. All outcomes are executable-oracle scores. No semantic label, human adjudication, agent-generated substitute label, or LLM-judge score enters the outcome. The registered primary estimand is the task-balanced paired difference $y_{Trajectory} - y_{Raw}$. The mandatory competing contrast is $Gain(Trajectory) - Gain(Generic)$, where $Gain(k) = y_k - y_{NoOp}$ within a checkpoint. No-op is the realized benefit and harm reference, not a third superiority estimand. Safety reports harm rate and mean negative change versus No-op for every supervisor condition. Secondary measures include abstention utility, returned bytes, rendered tokens, tool calls, latency, and total supervisor-plus-worker inference cost.

Repeated continuations estimate worker stochasticity but do not create independent task instances. We report task-level outcomes and paired uncertainty over held-out task-family clusters. Support for objective intervention utility requires Workspace Trajectory to satisfy preregistered decision rules for the primary Full Raw contrast, the mandatory Generic contrast, and intervention harm. Before execution, those rules must freeze interval/effect thresholds and the acceptable harm-rate and mean-negative-change margins. If Workspace Trajectory ties Full Raw while meeting the preregistered lower-retrieval-cost criterion, the result supports only an efficiency claim. A tie with Generic or No-op rejects the stronger claim.

RQ1: Objective Intervention Utility

RQ1 tests the primary Workspace Trajectory–Full Raw score contrast and the mandatory difference between Workspace Trajectory's and Generic's respective gains over No-op. No-op separately anchors realized benefit and harm. Completed evidence is limited to a mechanics-only pilot on Harness-Bench (Yao et al. 2026) revision `1025086a446653702b80cfb48babbeec35db6b2c`. The pilot used task `058-multiday-project-state` at its Day-2/Day-3 boundary. Two distinct prefix Codex sessions yielded 133 Raw records and seven normalized actions. The run verified immutable snapshots, fork manifests, worker arguments and environment, source parity, network denial, future-task and oracle non-leakage, and agreement between duplicate oracle invocations.

Realized pilot configuration. The worker used Codex CLI 0.144.6 with `gpt-5.6-sol` configured for medium reasoning. The supervisor was Qwen3.6-27B Q4_K_M at the frozen model hash reported by the artifact, served by llama.cpp revision `2d973636e292ee6f75fadcf08d29cb33511f509f` on one 32-GB RTX 5090. Each supervisor had a 65,536-token context, 2,048-token output cap, 16,384-token and 65,536-byte evidence caps, 2,048-token/8,192-byte response caps, 24 tool calls, and a 20-minute timeout. Rendered request sizes were 1,292, 1,737, and 2,076 tokens for Generic, Full Raw, and Workspace Trajectory. Elapsed supervisor times were 2.9, 2.3, and 2.6 seconds. Each condition executed one final worker continuation and two oracle invocations.

The three supervisor arms used the same model, prompt, and budget configuration. Full Raw and Workspace Trajectory also shared the source store but received their registered interfaces. All three supervisors made zero tool calls and cited no source IDs. Full Raw and Workspace Trajectory therefore failed the historical-evidence engagement gate. Generic's zero calls are reported but it had no historical-evidence tools. Scores were 0.8594 for No-op, Full Raw, and Workspace Trajectory, and 0.9219 for Generic. The run validates only checkpoint, fork, continuation, and oracle mechanics. It does not test Workspace Trajectory retrieval or objective intervention utility.

In separate fresh executions, the preregistered No-op headroom screen then evaluated six fixed tasks once. Its task-058 score was not reused from the pilot:

Task	Official score
057 interruption/resume	0.6154
058 multiday state	0.8594
059 event update/replan	1.0000
060 cancellation/cleanup	1.0000
103 policy update/replan	1.0000
105 partial-batch ledger	0.4994

Table 1: Registered Harness-Bench no-op headroom results. The gate required at least four of six tasks below 0.95, but only three qualified.

The failed gate prevented the four-condition RQ1 matrix. We also exclude post hoc selection of the three lower-scoring tasks. The pilot supplied mechanics-only evidence because Full Raw and Workspace Trajectory made zero evidence-tool calls. RQ1 therefore remains unanswered.

RQ2: Query Contribution

RQ2 asks whether removing artifact history, session differences, or action effects reduces objective intervention utility. Each representation ablation would remove one relation query while preserving complete prior-session source membership, Raw tools, checkpoint, and budgets. A separate matched source-scope contrast would remove earlier-session records identically from Full Raw and Workspace Trajectory. It is not a relation-query ablation. Because the four-condition RQ1 matrix was not admitted, the RQ2 gate did not open. RQ2 remains unanswered. Conditional on RQ1 support, the required evidence comprises three one-query removals followed by the matched earlier-session source-scope contrast.

RQ3: Generalization and Safe Supervision

RQ3 asks whether an RQ1-supported Workspace Trajectory advantage generalizes across held-out coding and scientific-work families, workers, and models under the preregistered harm rule. Because RQ1 is unanswered, RQ3 is closed. Workload qualification begins with structural properties, not treatment scores. For coding, SWE-ContextBench (Zhu et al. 2026b) is eligible only if related-task sequences retain one persistent workspace lineage and expose a genuine fresh-session boundary. Otherwise, a compatible SWE-INTERACT (Raghavendra et al. 2026) construction is required. For scientific work, CORE-Bench (Siegel et al. 2024) is eligible only if its official runner can pause at a structural boundary, continue in a fresh session, and apply the unchanged oracle.

Within each candidate domain, a preregistered manifest freezes structural eligibility and reserves disjoint RQ1 task-family clusters plus untouched RQ3 families. Coding and scientific work must separately pass structural qualification, a No-op headroom gate, and a mandatory historical-evidence engagement pilot before their four-condition RQ1 comparisons. Only after an admitted RQ1 matrix satisfies the support rule may its held-out RQ3 matrix run after the same three gates. Held-out low-scoring tasks cannot backfill RQ1. Future admitted runs will retain both oracle payloads and hashes. No task is selected from observed model scores.

An admitted RQ3 experiment must report qualification, task-family clusters, paired outcomes, harm, abstention, cost, and uncertainty. RQ3 remains unanswered.

Limitations. Native agent records may omit implicit reads, subprocess effects, or actions outside instrumented tools. Future evaluations require controlled source-coverage audits to quantify this gap. AgentSight system observations will form a separately named evidence ablation only if omission is material. Path-based identity remains imperfect when a tool copies content without an observed rename.

The supervisor message can change worker behavior for reasons unrelated to evidence quality. The Generic matched-inference control, unchanged prompts, byte-identical forks, and total-budget accounting reduce but do not eliminate this threat.

Objective evaluators can reward narrow benchmark behavior. Held-out task families and both coding and scientific-work domains are therefore required before broad claims.

The representation records actions and artifact effects without inferring intent or failure reasons and without judging whether repeated work was wasteful. A supervisor may later infer those semantics, but the current study evaluates only whether the supervisor's intervention improves an executable outcome.

Related Work

Trajectory diagnosis and process analysis. AgentDiagnose measures process competencies (Ou et al. 2025). AgentRx localizes failures (Barke et al. 2026). TrajAudit retrieves evidence from repository traces (Wang, Xie, and Huo 2026), and TRAJEVAL analyzes search, read, and edit stages (Kim et al. 2026). Accordingly, our claim excludes generic diagnosis, trace retrieval, and stage analysis. We instead test objective intervention utility from cross-session workspace relations against a same-source Full Raw interface.

Intervention and harness repair. HarnessFix maps failed trajectories to repairable harness mechanisms (Chen et al. 2026). AgentForesight audits prefixes for early failure (Zhang et al. 2026), and REFLECT uses controlled replay to validate attribution (Lin et al. 2026). Our contribution is not intervention itself. The matched checkpoint protocol is designed to test whether persistent-workspace evidence improves continuation beyond the Generic matched-inference condition and Full Raw access.

Long-lived agents and persistent workspaces. AgingBench studies reliability across many sessions (Zhu et al. 2026a). Cross-session threat detection shows that session-local guards miss aggregate behavior (Azarafrooz 2026), and plans can lose influence after context management (Mehta and Datta 2026). OR-Space makes persistent, multi-artifact workspaces part of the task (Zhou et al. 2026). These results motivate the setting but do not answer the matched Full Raw–Workspace Trajectory intervention question.

Ethical Considerations

Native traces can contain source code, prompts, paths, credentials, and private experimental data. Runtime credentials and session homes are not research artifacts and must be purged before publication. Any released trajectory must comply with the source workspace's applicable permissions. Aggregate metrics do not authorize publishing private records.

An automatic supervisor can cause harm by interrupting useful work. The protocol therefore includes abstention, no-op comparison, harm reporting, and bounded intervention rather than autonomous control without evaluation.

Conclusion

Long-horizon agents transform persistent workspaces across transient sessions. AGENT NEBULA exposes this process through a source-linked action trajectory and is designed to evaluate supervision by the executed outcome. The registered comparison tests Workspace Trajectory against Full Raw and Generic, with No-op anchoring benefit and harm. The pilot validated mechanics, but headroom blocked the RQ1 matrix. An objective cross-domain experiment remains necessary before claiming improved process-level oversight.

References

Azarafrooz, A. 2026. Cross-Session Threats in AI Agents: Benchmark, Evaluation, and Algorithms. arXiv:2604.21131.

Barke, S.; Goyal, A.; Khare, A.; Singh, A.; Nath, S.; and Bansal, C. 2026. AgentRx: Diagnosing AI Agent Failures from Execution Trajectories. arXiv:2602.02475.

Chen, M.; Wang, J.; Liu, Z.; Wang, Y.; Zheng, H.; and Wang, Q. 2026. From Failed Trajectories to Reliable LLM Agents: Diagnosing and Repairing Harness Flaws. arXiv:2606.06324.

Kim, M.; Wang, D.; Cui, S.; Farmahinifarahani, F.; Zhuo, T. Y.; Garg, S.; Ray, B.; Mukherjee, R.; and Kumar, V. 2026. Coherence Collapse: Diagnosing Why Code Agents Fail After Reaching the Right Code. arXiv:2603.24631.

Lin, X.; Wang, Y.; Kwok, T. S. T.; Guo, D.; Nale, S. A.; Fleming, C.; and Cheng, G. 2026. REFLECT: Intervention-Supported Error Attribution for Silent Failures in LLM Agent Traces. arXiv:2606.09071.

Mehta, A.; and Datta, A. 2026. Plans Don't Persist: Why Context Management Is Load Bearing for LLM Agents. arXiv:2606.22953.

Ou, T.; Guo, W.; Gandhi, A.; Neubig, G.; and Yue, X. 2025. AgentDiagnose: An Open Toolkit for Diagnosing LLM Agent Trajectories. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP): System Demonstrations*, 207–215.

Raghavendra, M.; Gunjal, A.; Sabharwal, A.; and He, Y. 2026. SWE-INTERACT: Reimagining SWE Benchmarks as User-Driven Long-Horizon Coding Sessions. arXiv:2606.30573.

Siegel, Z. S.; Kapoor, S.; Nadgir, N.; Stroebl, B.; and Narayanan, A. 2024. CORE-Bench: Fostering the Credibility of Published Research Through a Computational Reproducibility Agent Benchmark. *Trans. Mach. Learn. Res.*, 2024.

Wang, M.; Xie, X.; and Huo, Y. 2026. TrajAudit: Automated Failure Diagnosis for Agentic Coding Systems. arXiv:2605.26563.

Yao, Y.; Tan, X.; Liu, C.-H.; Li, Y.; Wang, Z.; Yu, W.; Tan, Z.; Tian, Y.; Zhao, G.; Sun, L.; Zhang, X.; and Yang, T. 2026. Harness-Bench: Measuring Harness Effects across Models in Realistic Agent Workflows. arXiv:2605.27922.

Zhang, B.; Zhu, J.; Shi, Z.; Liu, D.; and Tang, R. 2026. AgentForesight: Online Auditing for Early Failure Prediction in Multi-Agent Systems. arXiv:2605.08715.

Zhou, C.; Lu, X.; Zhao, J.; Lin, J.; Ge, D.; and Ye, Y. 2026. OR-Space: A Full-Lifecycle Workspace Benchmark for Industrial Optimization Agents. arXiv:2605.28158.

Zhu, J.; Ro, Y.; Robertson, J. T.; Wang, K.; Li, J.; Vikalo, H.; Akella, A.; and Wang, Z. 2026a. Your Agents Are Aging Too: Agent Lifespan Engineering for Deployed Systems. arXiv:2605.26302.

Zhu, J.; Wu, J.; Hu, M.; Zhu, S.; Pan, J.; Shen, W.; Yang, Y.; Liu, F.; Hao, J.; Jin, Y.; Ho, Q.; and Xu, M. 2026b. SWE Context Bench: A Benchmark for Context Learning in Coding. arXiv:2602.08316.